



BGP upgrade

Internet's security sucks, US wants to fix it

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There's a fundamental problem in the internet's fabric, of how it routes traffic from one point or network to another, that makes it less secure than what it should be. You don't think about it every single time when you fire up your favourite websites on your smartphone or computer's web browser, or generally whenever you access any information over the internet, but this design flaw of how the internet functions at its core remains. The weakest link in the internet's chain? BGP or Border Gateway Protocol

Despite being the foundational protocol governing internet routing, BGP has long been plagued by security vulnerabilities – which we'll get to later in the article. To address this critical issue, the White House Office of the National Cyber Director (ONCD) has released a BGP upgrade roadmap, which essentially calls for the widespread adoption of Resource Public Key Infrastructure (RPKI) to mitigate BGP's risks. Additionally, the US government is also establishing a

public-private working group to develop resources and materials to support the implementation of internet's routing security controls going forward.

What is BGP: How does it enable the internet to work?

First described as a networking routing technique as early as 1989, it wasn't until 1994 that BGP or Border Gateway Protocol became ingrained into the basic fabric of the internet as we know it. That makes BGP a 30-year-old technology, as far as the internet's concerned, with precious little upgrades. This is a problem, because BGP is responsible for determining the best path for data to travel across the internet. It helps routers exchange information about



reachable networks and routes traffic in an efficient manner.

According to the White House report, the internet is made up of nearly 74,000 independently operated but interconnected networks called Autonomous Systems (ASes) – these include everything from residential broadband to top-secret government networks, business and critical infrastructure enterprise, mobile wireless, cloud service and internet transit networks. One thing that all these networks have in common is the use of BGP to help exchange data and information.

The internet needs three key pieces to function – TCP/IP protocol, BGP, and DNS (Domain Name Server). Without getting into too many technicalities, think of the internet as a town – where TCP/IP is the underlying infrastructure of roads that connects various different sections of the town, DNS service is like a directory of all the names and locations of homes and business spread across the town, and BGP is like the GPS navigation that plans the most efficient route for you to travel from one location in the town to another.

Regardless of whether you connect to the internet from your phone or laptop (or any other device, for that matter), whenever you do something online you are essentially sending or receiving data. And when you type a website's address in your web browser and press enter or launch Amazon's shopping app on your phone to buy something instantly, data that's going out of your home / office / public network's router or gateway has several ways to reach its desired destination – which is to locate the website address you typed, Amazon or otherwise. Similarly, data that's coming to you from the internet has various paths it can choose to take. It boils down to BGP for finding the shortest, most efficient and therefore the quickest route for data coming and going to and from the internet.

Whether it's undersea inter-continental cables that carry 99% of the internet traffic around the world